EE 446: TinyML for Ultra Low-Power Edge Computing
Final Project Report Guidelines and Rubric

Department of Electrical and Computer Engineering
University of Washington

Total: 100 points
Due: June 13th, 11:59 PM via Canvas – No Extensions Allowed

Submission Requirements

- Submit one PDF report per team through Canvas.
- Use the filename format: `TeamName_EE446_FinalProjectReport.pdf`.
- Maximum length: **15 pages**, including title page, figures, tables, and references.
- The report must be self-contained. It should clearly explain the problem, dataset, models, TinyML deployment, results, and contributions of each team member.
- Include screenshots or photos showing on-device inference running on the target hardware. For this course, the expected target platform is the **Arduino Nano 33 BLE Sense Rev2** unless your team received approval to use another device.
- **Mandatory GitHub Repository Submission:** Along with the final report, each team must submit a **public GitHub repository link** containing the complete project implementation. The repository must include all relevant code, notebooks, trained models, deployment files, and artifacts used in the project, such as public Edge Impulse links, `.ipynb` files, model files such as `.pt`/`.h5`/`.tflite`, Arduino `.ino` files, preprocessing scripts, datasets or dataset links, and clear instructions for reproducing the results.
- **Important:** Failure to provide a working public repository link, submitting incomplete or partial code, or submitting code that does not run may result in **zero or significant loss of marks** for the final report. All results, tables, screenshots, claims, and deployment details presented in the report must match the submitted implementation.

Competitive Project Expectations

A competitive final project should go beyond the baseline implementation by including advanced technical components. The expected number of advanced components depends on team size.

Team Size	Competitive Project Expectation
1–2 students	At least 1 advanced component
3 students	At least 2 advanced components
4 students	At least 3 advanced components

Advanced components may include:

1. **Cascading model compression techniques:** knowledge distillation, pruning, quantization-aware training, post-training quantization, or combinations of these. Compare model size, accuracy, latency, and memory usage across configurations.
2. **Multiple models or modalities:** compare more than one model or approach, such as CNNs, DNNs, autoencoders, supervised models, unsupervised models, or tiny ensemble approaches. Even if only one model is deployed, the report should compare the alternatives experimentally.
3. **Fine-tuning using onboard-sensor data:** fine-tune at least one model using data collected from the target hardware or a realistic deployment setup, such as the Arduino microphone, IMU, camera, or other sensors. Compare original and fine-tuned performance.
4. **BLE or other communication methods for inference results:** transmit inference outputs using BLE, a mobile app, local dashboard, web API, attached display, or another communication method. This should go beyond only printing results to the Arduino IDE Serial Monitor.

Required Report Structure

1. Title Page

Include the project title, team name, team member names, course name, quarter, and a short one-sentence project summary.

2. Team Member Contributions [5 points]

Briefly describe each team member's contribution using two to three sentences per member. Be specific about work on data collection, model development, compression, deployment, demo integration, evaluation, writing, and debugging.

3. Introduction and Project Objective [8 points]

- Explain the real-world problem and why it is suitable for TinyML. *3 pts*
- Clearly state the project goal, input/output task, and target deployment setting. *3 pts*
- Define success criteria, such as accuracy, latency, memory usage, model size, energy relevance, or usability. *2 pts*

4. Dataset and Experimental Setup [12 points]

- Describe the dataset source, sensor type, classes/labels, number of samples, and any data collection process. *3 pts*
- Explain preprocessing, feature extraction, normalization, windowing, augmentation, and label preparation. *3 pts*
- Clearly report train/validation/test splits and justify how data leakage was avoided. *3 pts*
- If onboard-sensor data was collected, describe how it was collected and how it was used for testing or fine-tuning. *3 pts*

5. Model Development and Advanced Components [20 points]

- Describe all major "big" ML models trained offline, such as CNNs, DNNs, LSTMs, autoencoders, classical ML models, or other baselines. *4 pts*
- Describe all TinyML-ready models, including compressed, quantized, pruned, distilled, or architecture-reduced versions. *5 pts*
- Explain the selected advanced component(s) and how they were integrated into the project. *5 pts*
- Provide important training details: loss function, optimizer, learning rate, epochs, batch size, early stopping, and software tools. *3 pts*
- Justify why the final deployed model was selected over alternatives. *3 pts*

6. TinyML System Design and On-Device Deployment [20 points]

- Include a clear system diagram showing the full pipeline from sensor/input to inference output. *4 pts*
- Explain how the model was converted for deployment, including TensorFlow Lite/TFLite Micro, C array conversion, model header files, operator requirements, and Arduino integration. *5 pts*
- Report deployment constraints, including model size, tensor arena size, RAM/Flash usage, and any relevant latency measurement. *4 pts*
- Include screenshots or photos showing on-device inference results, such as Serial Monitor, Serial Plotter, BLE app, dashboard, display output, or logged inference results. *4 pts*
- If BLE or another communication method was used, explain the data format, transmitted values, and how inference results are displayed or consumed. *3 pts*

Recommended Results Tables

Your report should include tables similar to the following. Adjust the metrics to match your task.

Metric Selection and Justification

Task Type	Metrics Used	Justification
Classification / Regression / Anomaly Detection	–	Explain why these metrics are appropriate for your dataset, task, class balance, and project goal.

Big ML Model Performance

Model	Train	Test	Key Metrics	Size	Worked? Why?
Baseline CNN/DNN/etc.	–	–	Accuracy / F1 / MAE / RMSE / etc.	–	Explain what worked, what did not work, and why.
Alternative model	–	–	–	–	Explain why it was selected or rejected.

TinyML and Compressed Model Performance

Model	Method	Train	Test	Size/Latency	Notes
Pruned model	Pruning	–	–	–	Include train/test results because retraining is involved.
QAT model	QAT	–	–	–	Include train/test results because quantization-aware retraining is involved.
KD model	Knowledge Distillation	–	–	–	Include train/test results for the student model.
Int8 TFLite model	PTQ	N/A	–	–	PTQ is applied after training; test result is required.

Deployment Resource and Performance Summary

Hardware	Model/Flash	RAM/Arena	Latency	Output	Evidence
Arduino Nano 33 BLE Sense Rev2	–	–	–	Serial / BLE / Web API / CLI / Dashboard	Screenshot or photo included.

On-Device Inference Evidence for 10 Independent Instances

ID	Input	Expected	Predicted	Correct?	Evidence
1	–	–	–	Yes/No	Screenshot/photo reference.
2	–	–	–	Yes/No	Screenshot/photo reference.
8	–	–	– 4	Yes/No	Screenshot/photo
10	–	–	–	Yes/No	Screenshot/photo reference.

8. Challenges, Limitations, and Future Work [5 points]

Briefly discuss the main technical challenges, how your team addressed them, remaining limitations, and one or two realistic next steps. Strong reports distinguish between what was completed, what partially worked, and what did not work.

9. Writing Quality, Reproducibility, and References [5 points]

- The report should be concise, well organized, and technically clear. *2 pts*
- Include enough implementation detail for the main results to be reproducible. Mention code structure, libraries, hardware, and important configuration choices. *2 pts*
- Cite datasets, papers, libraries, tools, tutorials, and any reused code using a consistent citation style. *1 pt*

Summary Rubric

Category	Points
Team member contributions	5
Introduction and project objective	8
Dataset and experimental setup	12
Model development and advanced components	20
TinyML system design and on-device deployment	20
Results and analysis	25
Challenges, limitations, and future work	5
Writing quality, reproducibility, and references	5
Total	**100**

Final Notes

A strong EE 446 final report should not only show that a model works in Python. It should show the complete TinyML engineering process: training, testing, compression, conversion, deployment, on-device inference, and careful analysis of tradeoffs. The best reports will make it clear why the deployed model is suitable for the target hardware and application.